PulseIQ Phase 1 Foundation

GitHub: https://github.com/adityamergurwar/pulseiq
Dashboard: https://pulseiq-workforce.streamlit.app

PulseIQ brings workforce and finance analysis into one application. A repeatable data pipeline checks source files, organizes accepted records in a DuckDB warehouse, and supplies governed metrics to the dashboard, API, and monthly reporting outputs.

Project overview

People analytics	Finance analytics	Decision support
Headcount, hiring and exits, attendance and leave, attrition risk, team productivity, and department impact.	Payroll and labor cost, budget versus actual, profit and loss, expenses, receivables, and cash collection.	Forecasts and anomalies, executive scorecard, KPI builder, and data quality and lineage views.

Project components

29	28	12	5
CSV source feeds	governed metrics	dashboard pages	API metrics

The dashboard groups related measures into focused pages and supports department and month filters. Metric definitions are shared across the dashboard and API so the same measure keeps consistent meaning wherever it is viewed.

Example dashboard results

In the September 2026 workforce view, headcount is 441, with 17 hires and 3 exits for a net increase of 14; headcount is 3.3% above the prior month. Attendance views show approved leave peaking at 573 days in January 2026, late arrivals peaking at 7.5%, and overtime share peaking at 0.5% of payroll cost.

How the project works

The implementation connects source data to decision-ready views through four layers. Each step keeps the data traceable and reuses the same governed definitions.

Stage	What happens
1. Ingest and validate	The pipeline reads 29 source CSV feeds. Schema and data-contract rules check expected types, required values, and feed-specific formats. Invalid records are kept in the audit trail with their source row and reason.
2. Shape the warehouse	DuckDB organizes raw, clean, warehouse, metrics, and audit layers. Conformed dimensions and fact tables support joined analysis, including employee history, while refresh and quarantine records provide an audit trail.
3. Serve governed measures	Metrics M01-M28 define shared workforce and finance calculations. Twelve Streamlit pages present them through filters, charts, and short insights. The FastAPI service exposes five selected metrics with definitions, lineage, data cutoff, and privacy metadata.
4. Package and refresh	Monthly PDF and Excel builders create shareable snapshots; the workbook has nine sheets. Docker and Compose package the services. A GitHub Actions workflow is configured for monthly, manual, and source or pipeline change runs.

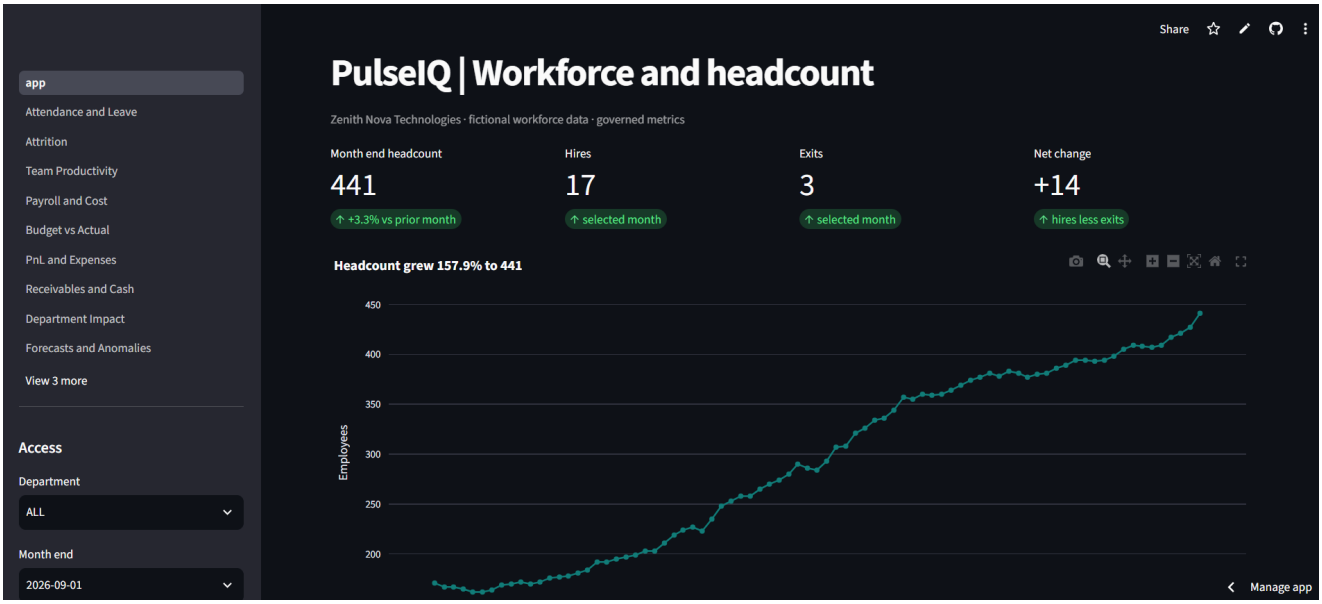
Validation results

Check	Result
Automated tests	13 passed.
ETL repeatability	29 source feeds processed consistently across two ETL runs.
Row disposition	62,770 input rows: 62,739 accepted and 31 quarantined with recorded reasons.
Metric layer	28 of 28 governed metric views returned results.
Payroll reconciliation	18 of 18 reference periods reconciled within INR 0.01.

Together, these checks show how the supplied snapshot moves through validation and into the same metric layer used for interactive analysis and reporting.

Dashboard screenshots

The two views below show the workforce summary and attendance analysis available in the app.



Workforce overview | September 2026: 441 employees, 17 hires, 3 exits and net growth of 14.



Attendance and Leave | Approved leave peaked at 573 days in January 2026; late arrivals reached 7.5% and overtime share 0.5%.